

Graph Centrality Analysis

Student Social Network - Neo4j GDS

1. Objective

This document presents the centrality analysis of the student friendship network. Four centrality measures are computed using the Neo4j Graph Data Science (GDS) library: Degree Centrality, PageRank, Betweenness Centrality, and Closeness Centrality. Each metric reveals a different dimension of a student's structural importance within the network.

2. GDS Graph Projection

Before running any algorithm, the friendship subgraph is projected into GDS memory as an undirected graph. This projection is reused by all four centrality algorithms.

```
CALL gds.graph.project(  
  'studentFriendshipGraph',  
  'Student',  
  { EST_AMI_DE: { orientation: 'UNDIRECTED' } }  
)  
  
YIELD graphName, nodeCount, relationshipCount;
```

3. Degree Centrality

Degree centrality measures the number of direct friendship connections each student has. Students with high degree are the most socially active and directly connected hubs of the network.

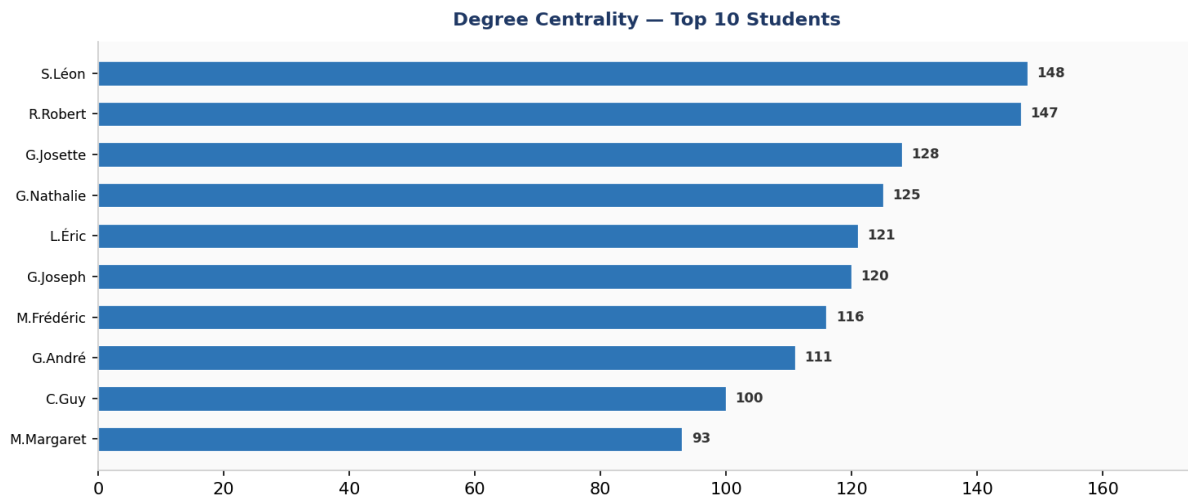
Cypher Query

```
CALL gds.degree.stream('studentFriendshipGraph')  
  
YIELD nodeId, score  
  
RETURN gds.util.asNode(nodeId).id AS studentId,  
       gds.util.asNode(nodeId).nom AS name,  
       score  
  
ORDER BY score DESC LIMIT 10;
```

Top 10 Results

Rank	Student ID	Name	Degree
1	6	Salmon Léon	148
2	10	Ribeiro Robert	147
3	7	Grégoire Josette	128
4	3	Guillot Nathalie	125

5	5	Leleu Éric	121
6	11	Gautier Joseph	120
7	9	Moreau Frédéric	116
8	33	Guichard André	111
9	8	Chauveau Guy	100
10	13	Mercier Margaret	93



4. PageRank

PageRank measures the influence of a student based not only on the number of connections but also on the importance of those connections. A student connected to many influential students will have a higher PageRank score.

Cypher Query

```
CALL gds.pageRank.stream('studentFriendshipGraph')
```

```
YIELD nodeId, score
```

```
RETURN gds.util.asNode(nodeId).id AS studentId,
```

```
gds.util.asNode(nodeId).nom AS name,
```

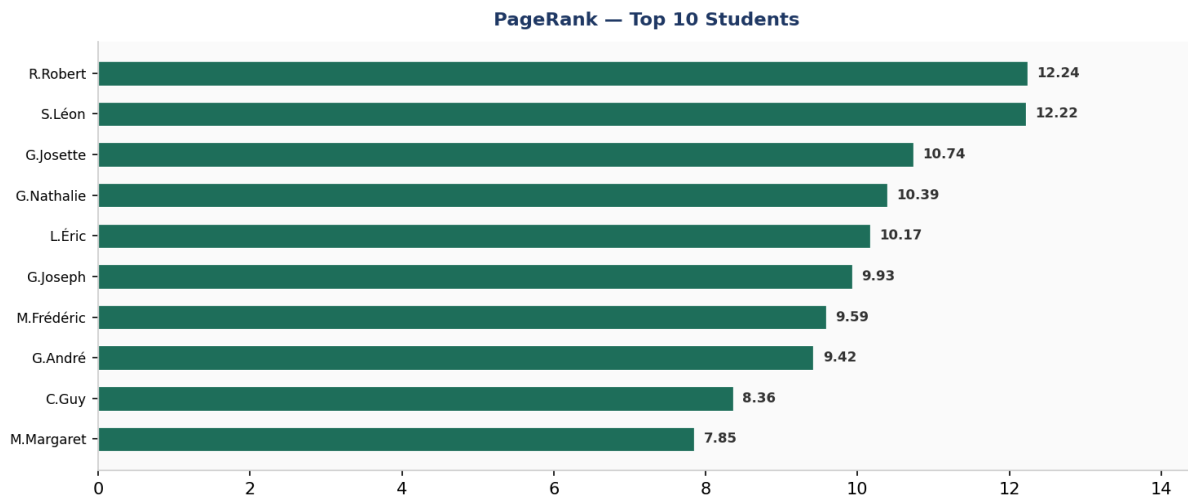
```
score
```

```
ORDER BY score DESC LIMIT 10;
```

Top 10 Results

Rank	Student ID	Name	PageRank Score
1	10	Ribeiro Robert	12.2400
2	6	Salmon Léon	12.2200
3	7	Grégoire Josette	10.7400
4	3	Guillot Nathalie	10.3900

5	5	Leleu Éric	10.1700
6	11	Gautier Joseph	9.9300
7	9	Moreau Frédéric	9.5900
8	33	Guichard André	9.4200
9	8	Chauveau Guy	8.3600
10	13	Mercier Margaret	7.8500



5. Betweenness Centrality

Betweenness centrality measures how often a student appears on the shortest path between other pairs of students. Students with high betweenness act as bridges between different parts of the network.

Cypher Query

```
CALL gds.betweenness.stream('studentFriendshipGraph')
```

```
YIELD nodeId, score
```

```
RETURN gds.util.asNode(nodeId).id AS studentId,
```

```
gds.util.asNode(nodeId).nom AS name,
```

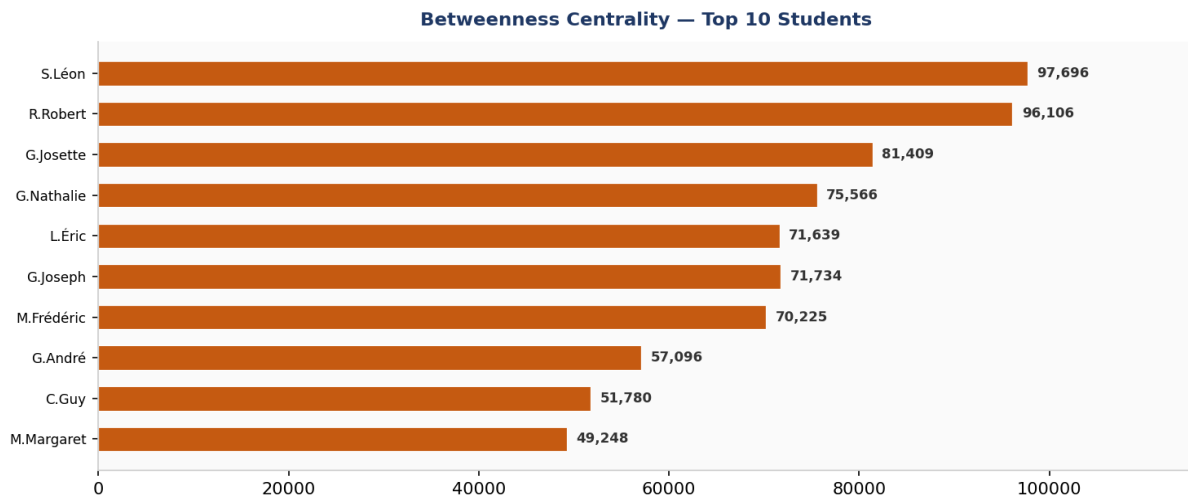
```
score
```

```
ORDER BY score DESC LIMIT 10;
```

Top 10 Results

Rank	Student ID	Name	Betweenness Score
1	6	Salmon Léon	97,696
2	10	Ribeiro Robert	96,106
3	7	Grégoire Josette	81,409
4	3	Guillot Nathalie	75,566

5	5	Leleu Éric	71,639
6	11	Gautier Joseph	71,734
7	9	Moreau Frédéric	70,225
8	33	Guichard André	57,096
9	8	Chauveau Guy	51,780
10	13	Mercier Margaret	49,248



6. Closeness Centrality

Closeness centrality measures how quickly a student can reach all other students in the network. A high closeness score means the student is on average closer to everyone else — making them efficient information spreaders.

Cypher Query

```
CALL gds.closeness.stream('studentFriendshipGraph')
```

```
YIELD nodeId, score
```

```
RETURN gds.util.asNode(nodeId).id AS studentId,
```

```
gds.util.asNode(nodeId).nom AS name,
```

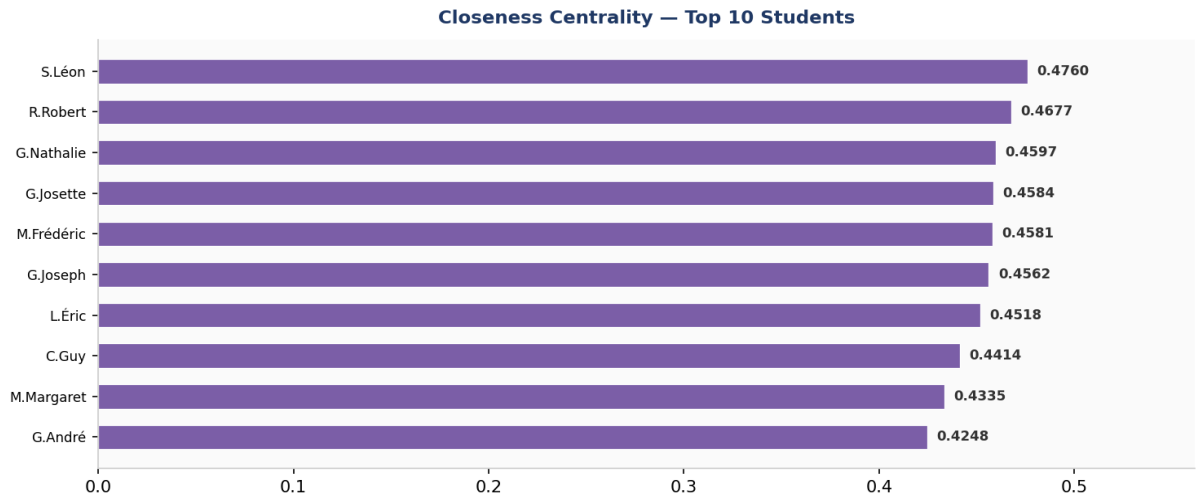
```
score
```

```
ORDER BY score DESC LIMIT 10;
```

Top 10 Results

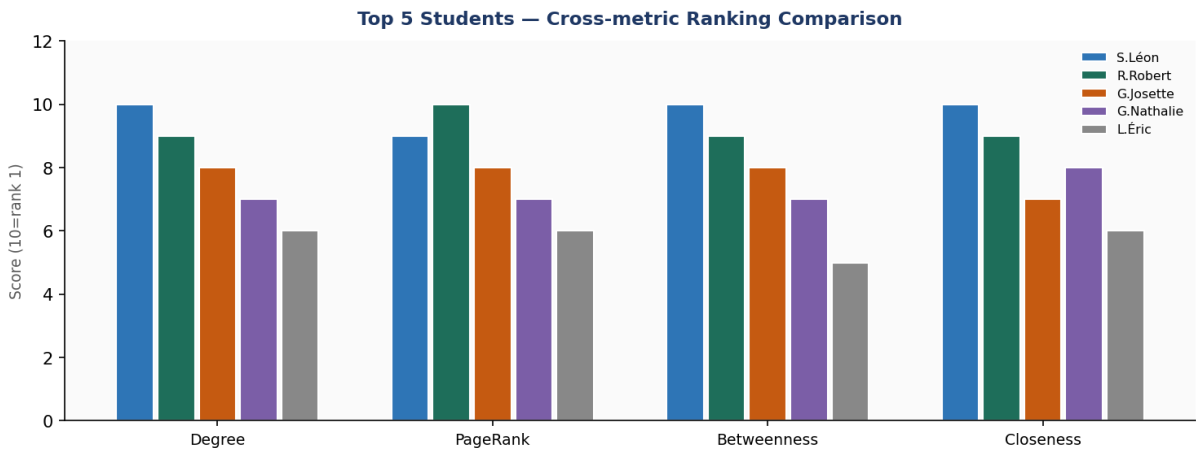
Rank	Student ID	Name	Closeness Score
1	6	Salmon Léon	0.4760
2	10	Ribeiro Robert	0.4677
3	3	Guillot Nathalie	0.4597
4	7	Grégoire Josette	0.4584

5	9	Moreau Frédéric	0.4581
6	11	Gautier Joseph	0.4562
7	5	Leleu Éric	0.4518
8	8	Chauveau Guy	0.4414
9	13	Mercier Margaret	0.4335
10	33	Guichard André	0.4248



7. Comparison of Rankings

The table below shows the rank of each student across all four centrality metrics. Students appearing consistently across all rankings are the most structurally important nodes in the network.



Student	Degree Rank	PageRank Rank	Betweenness Rank	Closeness Rank
Salmon Léon	1	2	1	1
Ribeiro Robert	2	1	2	2
Grégoire Josette	3	3	3	4
Guillot Nathalie	4	4	4	3

Leleu Éric	5	5	5	7
Gautier Joseph	6	6	6	6
Moreau Frédéric	7	7	7	5
Guichard André	8	8	8	10
Chauveau Guy	9	9	9	8
Mercier Margaret	10	10	10	9

8. Analysis of Differences

Students appearing in all four Top 10 rankings

All 10 students above appear in every ranking simultaneously. This is a strong indicator that the top hubs of this network are consistently central regardless of the metric used, suggesting a highly concentrated influence structure.

Why Degree and PageRank rankings differ

Ribeiro Robert ranks 2nd in Degree but 1st in PageRank, while Salmon Léon ranks 1st in Degree but 2nd in PageRank. This means Ribeiro Robert's 147 friends are on average more influential than Salmon Léon's 148 friends. PageRank rewards quality of connections, not just quantity.

Why some students have high Betweenness

Salmon Léon and Ribeiro Robert have the highest betweenness scores (97,696 and 96,106), meaning they appear most frequently on shortest paths between other student pairs. They act as bridges between different social clusters — removing them would significantly fragment the network's connectivity.

Why some students have high Closeness

Salmon Léon has the highest closeness score (0.476), meaning they can reach any other student in the fewest hops on average. Guichard André ranks 8th in Degree but drops to 10th in Closeness, suggesting their connections are more locally clustered rather than distributed across the network.

9. Conclusion

The centrality analysis reveals a consistent hierarchy of influential students in the friendship network. Salmon Léon and Ribeiro Robert emerge as the most central students across all four metrics, confirming their role as key connectors of the network. The strong correlation between Degree, PageRank, and Betweenness rankings suggests that in this synthetic network, high connectivity directly translates to high influence and high bridging potential. These centrality scores will be used as features in the community detection and recommendation modules to improve result quality.